

Mohamed Firas Ben Khalifa
3rd Year Engineering Student – Data Science Specialization
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PROFILE

Data Science and AI engineering student (Bac+5) at ESPRIT, building and deploying AI-driven solutions in data-focused environments. Hands-on experience in machine learning, deep learning, and LLM applications. Looking for a six-month internship to build agentic AI and data-driven products within an ambitious team.

PROFESSIONAL EXPERIENCE

Proxym – IT Engineering Company Intern 8-Week Internship	Sousse, Tunisia 2026
- Designing a multi-agent support ticket platform (Angular, Spring Boot, PostgreSQL, FastAPI) with a hybrid ML + LLM triage pipeline routed by confidence score. - Architected asynchronous ticket ingestion using a message queue for high volume imports, with vector search for duplicate detection and similarity matching. - Built a RAG resolution agent that drafts cited response suggestions with mandatory human validation before anything is sent, alongside a secured Text to SQL agent restricted to read only views with strict query validation. - Set up LLM observability and an evaluation harness tracking precision, recall, and retrieval quality. Technologies: RAG, LangGraph, LLM Integration, Text-to-SQL, FastAPI, pgvector, RabbitMQ, PostgreSQL, Spring Boot, Angular, Docker, GitHub	
Whitecape Technologies – IT Engineering Company Intern 6-Week Internship	Sousse, Tunisia 2025
- Designed and built a full-stack intelligent notepad web application (Spring Boot + React + MySQL) with a REST API for data management. - Integrated and deployed a local LLM (TinyLlama) via Flask for automatic content suggestion, reducing writing time by 70%. - Applied OAuth authentication with Google and GitHub for secure user access. - Containerized the entire application using Docker for streamlined deployment. - Set up an automated emailing and SMS notification system. Technologies: Spring Boot, React.js, MySQL, REST API, OAuth, Docker, Flask, TinyLlama	

PROJECTS

SkillBridge AI — AI Copilot for Recruitment & Education - Built a full-stack AI copilot that extracts skills from resumes, computes explainable match scores, and generates personalized learning paths. - Developed an HR agent that retrieves salary benchmarks and market signals, paired with a rubric-scored mock interview simulator. - Integrated a persistent AI Career Twin aggregating matching results, interview sessions, and learning plans into an evolving profile. Technologies: Python, FastAPI, React, TypeScript, LangChain, LLMs
AgriSmart — Autonomous Mobile-First Agricultural Platform - Built a mobile application integrating AI agents orchestrated by LangChain to automate irrigation recommendations. - Implemented offline plant and animal disease detection via Computer Vision (PyTorch / TensorFlow Lite) with real-time inference. - Enhanced crop monitoring with satellite analysis (NDVI, VRA Mapping) and GPS-based livestock management. Technologies: Flutter, FastAPI, PyTorch, TensorFlow Lite, LangChain, Computer Vision, GPS
AIGIS — Multimodal AI Assistant for Structural Diagnostics - Designed a multimodal AI assistant combining a trained CNN for real-time structural damage detection with a full RAG pipeline for automated diagnostic report generation. - Deployed the full-stack application (FastAPI/React) to production, integrating LLMs for contextualized, domain-specific responses. - Coordinated technical and functional stakeholders across the full project lifecycle. Technologies: Python, CNN, Deep Learning, Computer Vision, FastAPI, React, RAG, LLMs, HuggingFace, Docker

EDUCATION

COMPUTER ENGINEERING CYCLE École Supérieure Privée d'Ingénierie et de Technologies (ESPRIT) BACCALAUREATE IN TECHNICAL SCIENCES Lycée Les Elites	2022 – Present Ariana, Tunisia 2022 Sousse, Tunisia
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Certifications

- Building Agentic AI Applications with LLMs – NVIDIA - Building LLM Applications with Prompt Engineering – NVIDIA - Rapid Application Development with LLMs – NVIDIA - Applications of AI for Anomaly Detection – NVIDIA	- Building RAG Agents with LLMs - NVIDIA - Building Transformer-Based Natural Language Processing Applications – NVIDIA - Fundamentals of Deep Learning – NVIDIA
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SKILLS

AI & Data Science: Agentic AI, RAG, LLMs, Prompt Engineering, Computer Vision, NLP, Deep Learning, Machine Learning, Anomaly Detection Frameworks: PyTorch, TensorFlow/TF Lite, HuggingFace Transformers, LangChain, scikit-learn, Flutter, Spring Boot, React Languages & Analysis: Python, R, SQL, PL/SQL, Java, C, C++, C#, PHP, HTML, JavaScript Tools & MLOps: Jupyter Notebook, Google Colab, Docker, GitHub, REST API, OAuth Databases: MySQL, MongoDB, SQL Server
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INTERESTS

Open-Source Contribution • Discovering new cultures • Learning new languages

LANGUAGES

Arabic (native) • French (C1) • English (C1) • German (B1)